

湖北大学 2025-2026 学年度第 2 学期课程考查试题纸

The paper of course exam

Name: Big Data Analysis and Applications

Content: Course Report

Teacher: Li Jie

Digital Lifestyle Analysis Course Report

High-Risk Screening, Digital Dependence Prediction, and Lifestyle Profiling

Student ID: _____ Name: _____

College: _____ Major and Grade: _____

Institution: _____ Grade and major: _____

Teacher's comments:

Total score: _____ Grading teacher: _____

Abstract

This report is prepared for the course Big Data Analysis and Applications and follows the teacher's seven grading dimensions: independent dataset selection, independent research topic design, data preprocessing, exploratory analysis and visualization, machine learning modeling and evaluation, report/code organization, and final reflection. The study uses the 2025 Digital Lifestyle Benchmark Dataset, a CSV structured tabular dataset with 3,500 records and 24 fields. The dataset is a synthetic educational and benchmark dataset, so all findings are used only for course modeling practice and behavioral analysis, not for medical diagnosis, individual health judgment, or causal inference.

The research topic is high-risk identification, digital dependence prediction, and user profiling in digital lifestyles. Three machine learning task types are implemented, which exceeds the course requirement of at least two types: classification for high_risk_flag screening, regression for digital_dependence_score prediction, and clustering for exploratory digital lifestyle profiling. The preprocessing stage includes data cleaning, feature extraction, PCA extension, LCA/GMM discussion, and strict target leakage control. Exploratory analysis uses descriptive statistics, correlation analysis, group comparisons, and visualizations such as bar charts, histograms, boxplots, scatter plots, heatmaps, ROC/PR curves, elbow curves, silhouette curves, PCA plots, and cluster profile heatmaps.

The validated experimental results are kept unchanged. For high-risk screening, the final classifier is Gradient Boosting with threshold=0.14, achieving Recall=0.6420, F1=0.5355, and PR-AUC=0.5084 on the test set. This is interpreted as a recall-oriented screening strategy rather than a diagnosis tool. For regression, digital_dependence_score is the main target, with $R^2=0.9839$, MSE=3.1471, and MAE=0.9982. By contrast, productivity_score obtains $R^2=-0.0041$ and is reported as a weak-prediction or negative-result analysis. For clustering, KMeans with k=3 obtains Silhouette=0.1860, so the clustering result is used only for exploratory user profiling instead of claiming naturally clear population boundaries.

Keywords: digital lifestyle; high-risk screening; digital dependence prediction; machine learning; clustering profile; big data analysis

Table of Contents

Abstract	2
Table of Contents	3
Chapter 1 Dataset Selection	4
1.1 Course Requirements for Dataset Selection	4
1.2 Candidate Dataset Comparison and Reasons for Excluding pfm_train / pfm_test.....	4
1.3 Final Dataset Source and Compliance Check	5
1.4 Feature Fields, Sample Scale, and Supported Tasks.....	5
Chapter 2 Independent Research Theme and Analytical Perspective.....	6
Chapter 3 Data Preprocessing	8
Chapter 4 Exploratory Analysis and Data Visualization.....	13
Chapter 5 Machine Learning Modeling, Hyperparameter Tuning, and Evaluation	20
Chapter 6 Report Structure and Code Presentation	31
Chapter 7 Conclusions and Personal Reflection	35
References.....	38
Appendix A Core Code	39
Appendix B Full Experimental Running Instructions.....	41
Appendix C Result Files, Figure List, and Static Showcase.....	42
Appendix D Personal Contribution and Academic Integrity Statement.....	44

Chapter 1 Dataset Selection

1.1 Course Requirements for Dataset Selection

This chapter directly corresponds to the first grading dimension: independent dataset selection. According to the course requirements, the dataset should come from a qualified public data platform such as Kaggle, Hugging Face, Alibaba Tianchi, Heywhale, or OpenML. It should be released or updated in 2024-2025, stored as a CSV structured table, contain at least 1,000 samples, contain at least 8 effective features, and naturally support at least two machine learning task types among regression, classification, and clustering.

The selected dataset must not merely be convenient. It should allow meaningful preprocessing, statistical analysis, visualization, machine learning modeling, metric evaluation, and code reproduction. Therefore, dataset selection is treated as a methodological decision rather than a file-loading step.

1.2 Candidate Dataset Comparison and Reasons for Excluding pfm_train / pfm_test

The pfm_train.csv and pfm_test.csv files were considered as candidates but were not used as the final dataset. Their source platform and release/update year were less stable for the final course requirement, and the employee attrition topic was highly similar to a previous experiment. In addition, the pfm files mainly support a classification-style task, while the regression task is less natural. To avoid repeating the earlier experiment and to build a richer final report, they were excluded from the main analysis.

1.3 Final Dataset Source and Compliance Check

The final dataset is the 2025 Digital Lifestyle Benchmark Dataset. It is suitable for this report because it contains digital behavior, lifestyle habits, mental-risk indicators, high_risk_flag, productivity_score, and digital_dependence_score. These variables support classification, regression, and clustering in one coherent analytical scenario.

Table 1-1 Dataset compliance summary against course requirements

Course requirement	Digital Lifestyle Benchmark Dataset (2025) status	Status
Source platform	Public Hugging Face dataset page	Meets
Release/update year	2025 dataset	Meets
File format	CSV structured tabular data	Meets
Sample size	3,500 rows, meeting the requirement of at least 1,000 rows	Meets
Number of effective features	24 fields, meeting the requirement of at least 8 effective features	Meets
Task support	Supports high_risk_flag classification, digital_dependence_score regression, and lifestyle clustering	Meets
Data nature	Synthetic educational/benchmark data; conclusions must be interpreted	Meets course modeling purpose

	cautiously and not used for medical diagnosis	
--	---	--

Table 1-1 shows that the dataset satisfies the formal course requirements. The dataset is synthetic educational/benchmark data, so it is appropriate for modeling practice but should not be interpreted as a medical or real-world diagnostic dataset.

1.4 Feature Fields, Sample Scale, and Supported Tasks

The dataset contains 3,500 records and 24 fields. The behavioral fields include variables such as device_hours_per_day, phone_unlocks, notifications_per_day, social_media_mins, and study_mins. Lifestyle fields include sleep_hours, sleep_quality, and physical_activity_days. Outcome fields include high_risk_flag, productivity_score, and digital_dependence_score.

This field design supports three task types: high_risk_flag classification, digital_dependence_score regression, and exploratory lifestyle clustering. The report therefore exceeds the course requirement of implementing at least two machine learning task types.

Chapter 2 Independent Research Theme and Analytical Perspective

2.1 Business Background and Problem Motivation

Mobile devices, social media platforms, frequent notifications, sleep habits, study time, and physical activity have become important components of daily digital lifestyles. A big data analysis report can use these observable behavior variables to explore digital well-being patterns in a structured and reproducible way.

The purpose of this report is not to diagnose health conditions, but to build a course-level modeling workflow that connects data quality, mathematical statistical analysis, visualization, supervised learning, and unsupervised profiling.

2.2 Report Title and Research Objectives

The report title is: High-Risk Identification, Digital Dependence Prediction, and User Profiling in Digital Lifestyles: A Classification, Regression, and Clustering Study Based on the 2025 Digital Lifestyle Benchmark Dataset.

Identify high-risk digital lifestyle samples using `high_risk_flag` as the classification target.

Predict `digital_dependence_score` as the main regression target.

Test whether `productivity_score` can be stably predicted by the current observable features.

Construct exploratory digital lifestyle user profiles through clustering.

2.3 Analytical Perspective and Research Questions

The analysis is organized from four perspectives. The behavioral perspective includes device time, notifications, phone unlocks, social media time, and study time. The lifestyle perspective includes sleep duration, sleep quality, and physical activity. The outcome perspective includes high-risk labels, digital dependence, and productivity. The methodological perspective includes classification, regression, and clustering.

The main research questions are: Can high-risk digital lifestyles be screened with leakage-controlled features? Can digital dependence be predicted from digital behavior and lifestyle habits? Is `productivity_score` predictable under the same feature constraints? Can clustering form interpretable digital lifestyle profiles?

2.4 Main Contributions and Technical Workflow

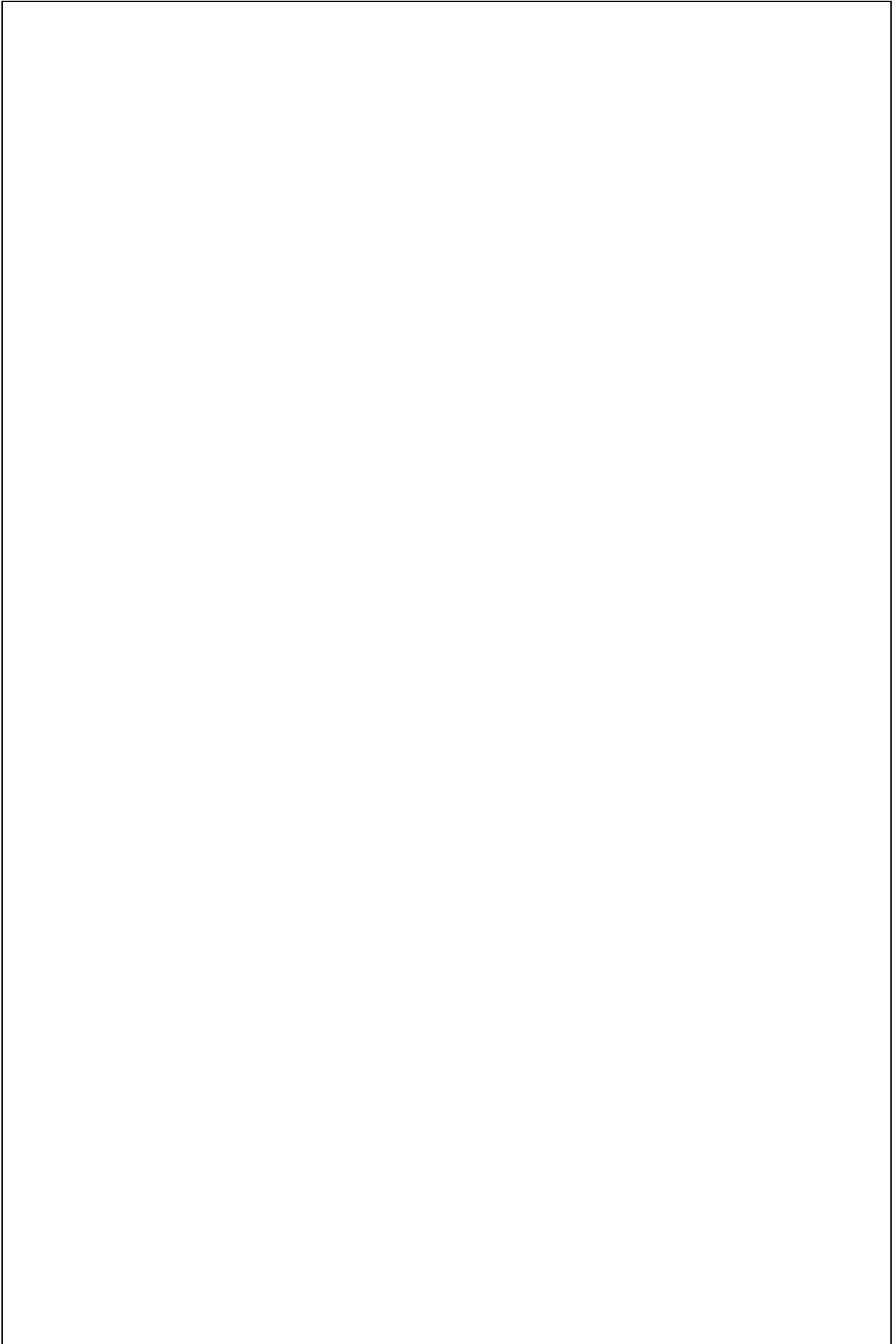
Completed independent dataset screening and explained why the pfm dataset was not selected.

Performed data cleaning, feature extraction, PCA extension, LCA/GMM discussion, and target leakage control.

Used statistical analysis and visualization to examine distributions, correlations, group differences, and risk-rate patterns.

Built classification, regression, and clustering tasks with tuning, metrics, and visualized evaluation evidence.

Organized notebooks, source code, result CSV files, figures, and the final Word report into a reproducible project structure.



Chapter 3 Data Preprocessing

3.1 Raw Data Loading and Field Inspection

The raw CSV file was loaded with pandas. The first checks verified the row count, column count, field names, and data types. The raw dataset contains 3,500 records and 24 fields. These checks ensure that subsequent preprocessing and modeling are performed on the expected structured table.

3.2 Missing Values, Duplicate Records, and Abnormal Data Checking

The preprocessing stage includes missing value checking, duplicate record checking, abnormal value checking, data rationality verification, and task-specific feature selection. The total number of missing values is 0, the number of duplicate rows is 0, and the number of duplicate id values is 0. Since no severe abnormal records were found, no sample deletion was performed.

Table 3-1 Compact preprocessing quality check

Check item	Check result	Processing strategy
Original sample size	3500	No records removed
Original number of fields	24	No fields removed
Total missing values	0	No deletion needed; the modeling pipeline still keeps missing-value handling capability
Duplicate rows	0	No deletion needed
Duplicate id count	0	No deletion needed
Rational range of key numerical fields	All passed	No records requiring deletion were found after checking
Engineered feature generation	All 7 engineered features generated	Used to supplement behavioral ratio and interaction information
Classification leakage-column exclusion	Passed	Outcome fields and id were excluded
Regression outcome control	Passed	high_risk_flag and other outcome variables were excluded
Clustering input control	Passed	Only behavioral and lifestyle numerical features were used

Table 3-1 summarizes the data cleaning evidence. The result supports the decision to keep the full sample while still preserving robust preprocessing operations in the modeling pipeline.

3.3 Data Rationality Verification

The rationality check covers age, device_hours_per_day, phone_unlocks, notifications_per_day, social_media_mins, study_mins, physical_activity_days, sleep_hours, sleep_quality, productivity_score, and digital_dependence_score. The check focuses on whether the observed minimum and maximum values are

consistent with reasonable behavioral ranges.

3.4 Feature Engineering and Feature Extraction

Feature extraction was conducted by constructing behavioral ratio and interaction features. The engineered features include social_media_hours, study_hours, notifications_per_device_hour, unlocks_per_device_hour, device_to_sleep_ratio, activity_sleep_interaction, and social_to_study_ratio.

Code 3-1 Behavioral feature extraction

```
def add_engineered_features(df):  
    out = df.copy()  
  
    out["social_media_hours"] = out["social_media_mins"] / 60  
  
    out["study_hours"] = out["study_mins"] / 60  
  
    out["notifications_per_device_hour"] = (  
        out["notifications_per_day"] / (out["device_hours_per_day"] + 1e-6)  
    )  
  
    out["unlocks_per_device_hour"] = (  
        out["phone_unlocks"] / (out["device_hours_per_day"] + 1e-6)  
    )  
  
    out["device_to_sleep_ratio"] = (  
        out["device_hours_per_day"] / (out["sleep_hours"] + 1e-6)  
    )  
  
    out["activity_sleep_interaction"] = (  
        out["physical_activity_days"] * out["sleep_quality"]  
    )  
  
    out["social_to_study_ratio"] = (  
        out["social_media_mins"] / (out["study_mins"] + 1)  
    )  
  
    return out
```

3.5 PCA Extension and LCA/GMM Discussion

PCA was used for dimensionality reduction visualization and auxiliary interpretation of clustering structure. The first two principal components explain about 42.41% of the total variance. PCA was not used as input for classification or regression and should not be interpreted causally.

Table 3-2 PCA explained variance table for the first 10 components

Principal component	Explained variance ratio	Cumulative explained variance
PC1	0.2694	0.2694

PC2	0.1547	0.4241
PC3	0.1366	0.5607
PC4	0.1258	0.6865
PC5	0.1030	0.7895
PC6	0.0759	0.8654
PC7	0.0430	0.9084
PC8	0.0425	0.9509
PC9	0.0274	0.9783
PC10	0.0152	0.9935

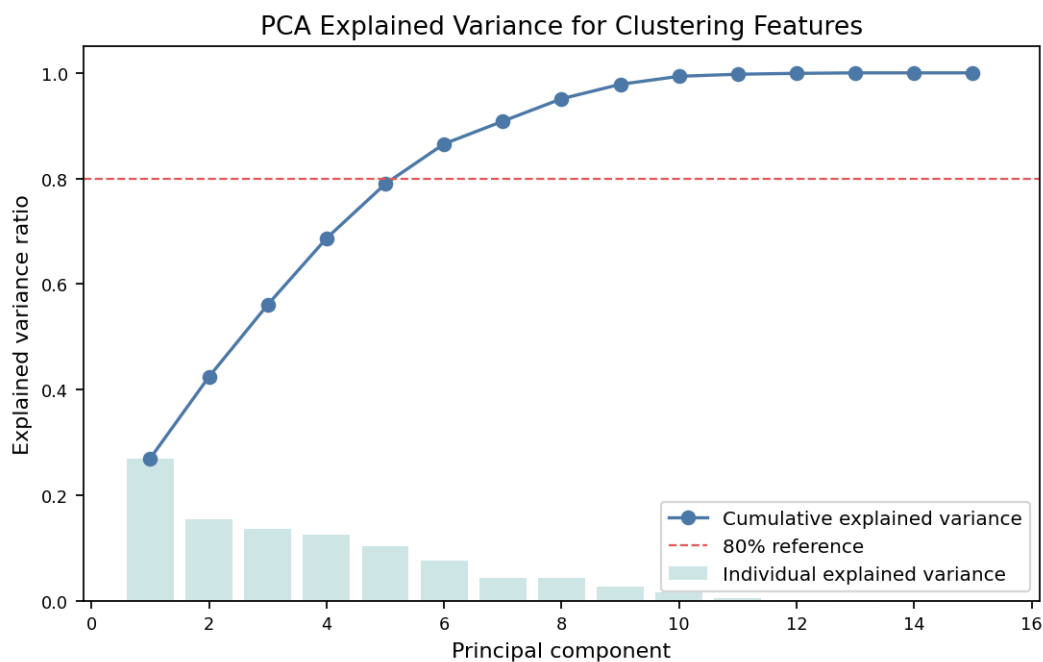


Figure 3-1 Cumulative explained variance of PCA components

The PCA curve provides visual evidence for the amount of information captured by the leading components. It supports dimensionality-reduction visualization, but it does not prove that the behavioral variables cause the outcome variables.

A separate LCA model was not introduced because the clustering inputs are mainly continuous behavioral and lifestyle variables. GaussianMixture was used as a probabilistic clustering method to approximate latent group structure and to complement KMeans and AgglomerativeClustering.

3.6 Feature Selection and Target Leakage Control

Target leakage control is essential because `high_risk_flag` is closely related to several mental-status and outcome variables. For the classification task, `anxiety_score`, `depression_score`, `stress_level`, `happiness_score`, `focus_score`, `productivity_score`, `digital_dependence_score`, `id`, and `high_risk_flag` were excluded from the input features.

For regression, `high_risk_flag` and other outcome fields were excluded. For clustering, only digital behavior and lifestyle numerical features were used as clustering inputs, while demographic and outcome variables were kept only for post-clustering interpretation.

Code 3-2 Task-specific leakage control

```
HIGH_RISK_LEAKAGE_COLUMNS = [
    "anxiety_score", "depression_score", "stress_level",
    "happiness_score", "focus_score", "productivity_score",
    "digital_dependence_score",
]

classification_drop = [
    "id", "high_risk_flag", *HIGH_RISK_LEAKAGE_COLUMNS
]

regression_drop = [
    "id", "high_risk_flag", "anxiety_score", "depression_score",
    "stress_level", "happiness_score", "focus_score",
]

clustering_features = [
    "device_hours_per_day", "phone_unlocks", "notifications_per_day",
    "social_media_mins", "study_mins", "physical_activity_days",
    "sleep_hours", "sleep_quality", "social_media_hours",
    "study_hours", "notifications_per_device_hour",
    "unlocks_per_device_hour", "device_to_sleep_ratio",
    "activity_sleep_interaction", "social_to_study_ratio",
]
```

3.7 Summary of Preprocessing Results

The preprocessing stage produced a clean and reproducible modeling table. No missing values, duplicate rows, or duplicate ids were found. Seven engineered features were generated successfully. The feature sets for classification, regression, and clustering were controlled separately to reduce leakage and keep the interpretation aligned with each task.

Chapter 4 Exploratory Analysis and Data Visualization

4.1 Descriptive Statistical Analysis

Before modeling, descriptive statistics, distributions, correlations, group comparisons, and risk-rate summaries were used to explore the mathematical structure of the dataset. The report examines mean, range, distribution shape, correlation, and group difference. This step helps determine whether later modeling results are plausible and interpretable.

4.2 Target Variable Distribution

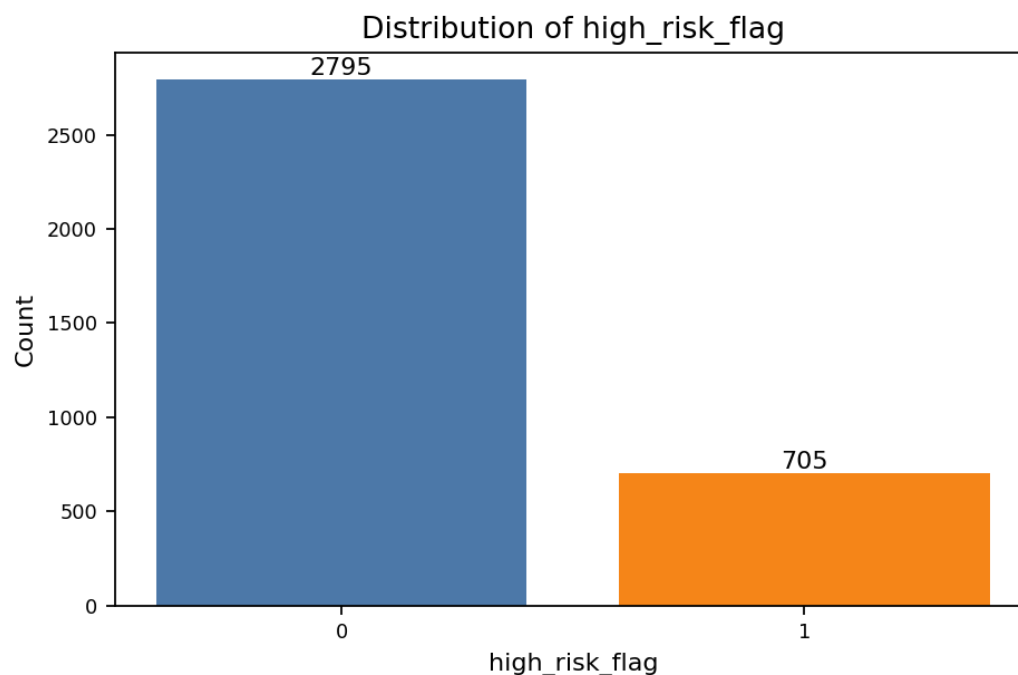


Figure 4-1 Distribution of high_risk_flag

The bar chart shows the class distribution of high_risk_flag. Because the positive high-risk group is not the majority class, Accuracy alone is not sufficient for model evaluation. This observation motivates using Recall, F1, PR-AUC, and the confusion matrix in Chapter 5.

4.3 Numerical Feature Distribution and Correlation Analysis

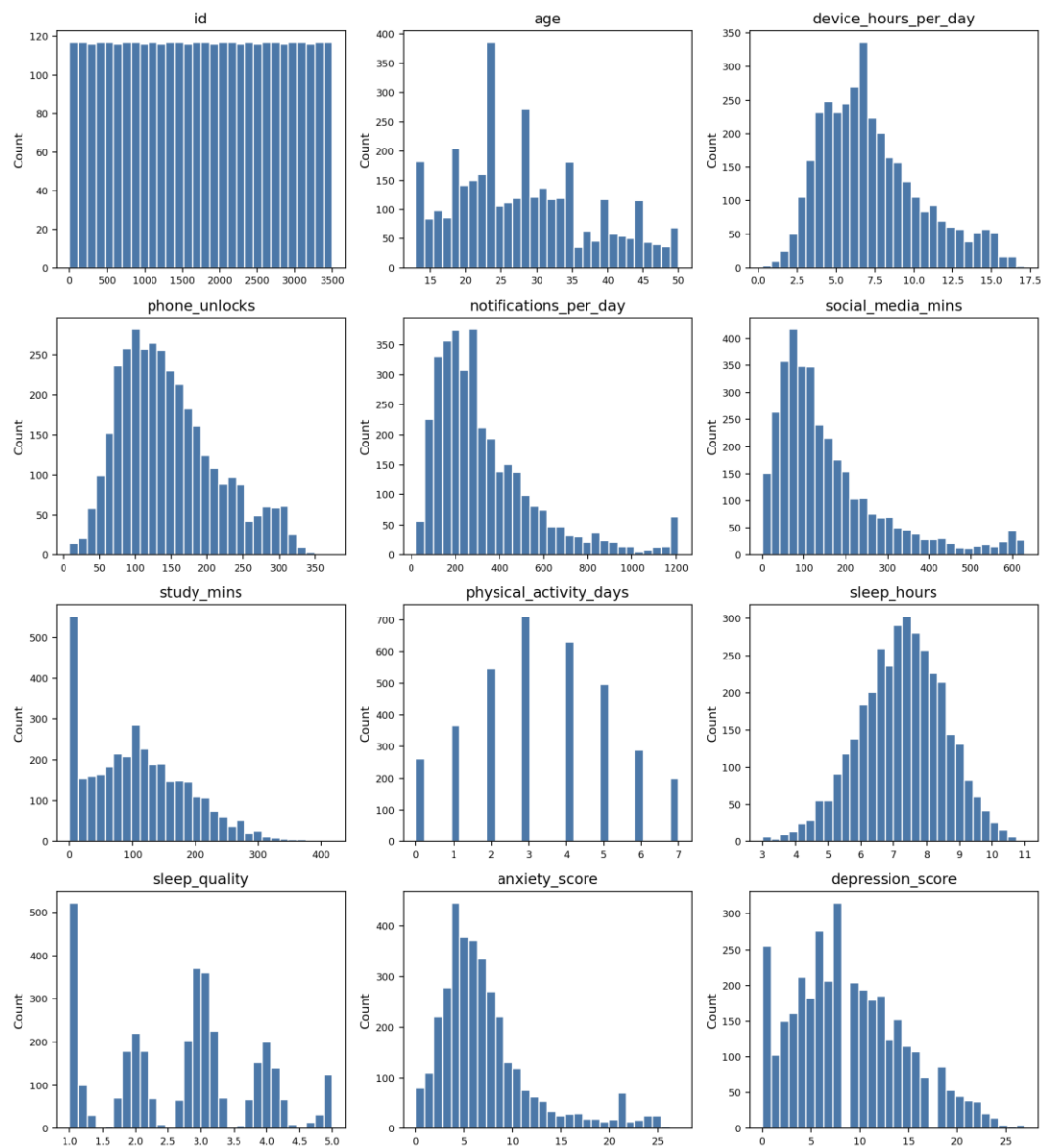


Figure 4-2 Distributions of core numerical variables

The histogram panel visualizes the distributions of core numerical variables such as device time, notifications, unlocks, sleep, digital dependence, and productivity. These distributions provide the mathematical background for deciding whether scaling, robust metrics, or nonlinear models may be useful.

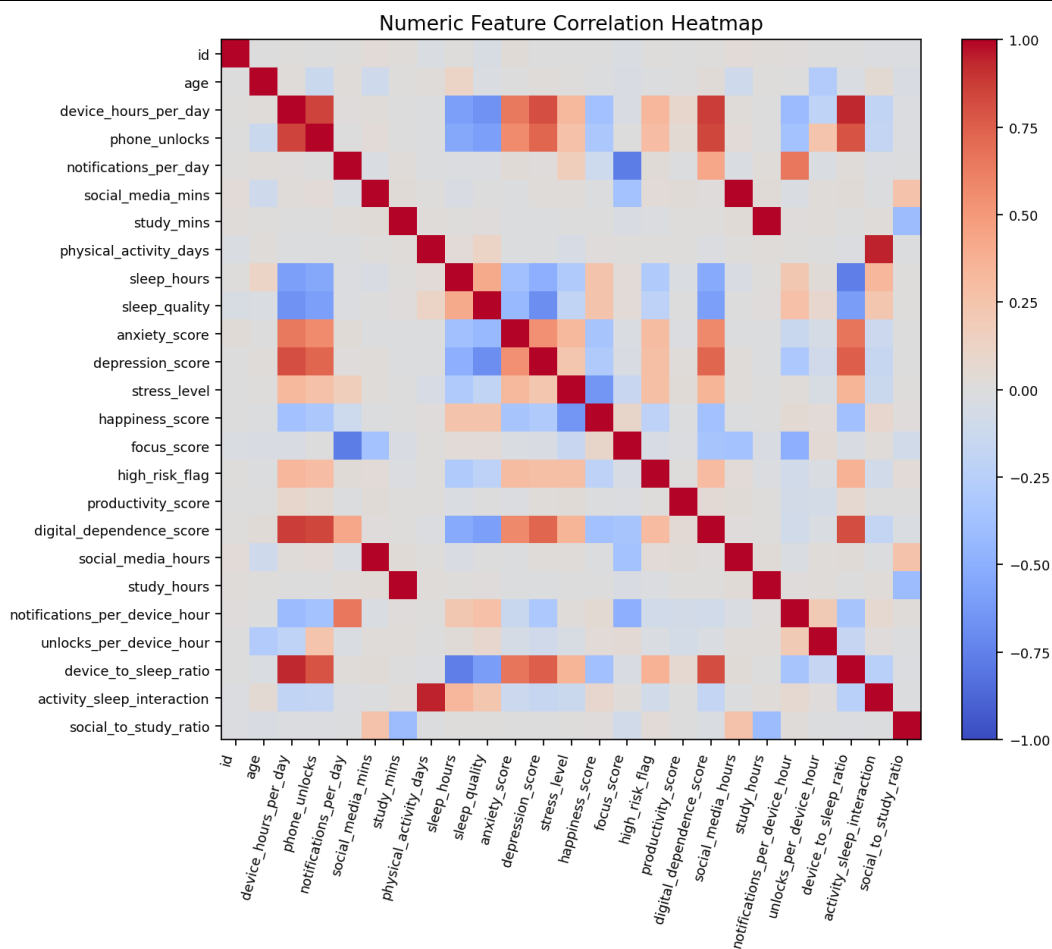


Figure 4-3 Correlation heatmap of numerical variables

The heatmap summarizes pairwise correlations among numerical variables. It helps identify associations between digital behavior variables and outcome variables, but correlation does not imply causation. The heatmap is used as exploratory evidence rather than causal proof.

4.4 Group Differences between High-Risk and Non-High-Risk Samples

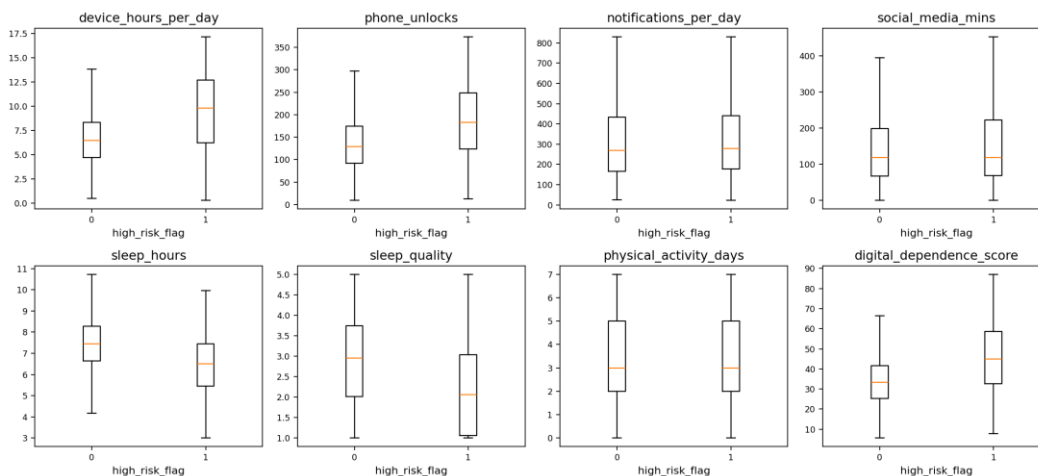


Figure 4-4 Behavioral differences by high-risk group

The boxplots compare behavioral and lifestyle variables between high-risk and non-high-risk samples. The

visual comparison supports later classification modeling by showing that several observable behaviors differ across risk groups. However, the figure should still be interpreted as descriptive group comparison.

4.5 Categorical Risk Rate Analysis

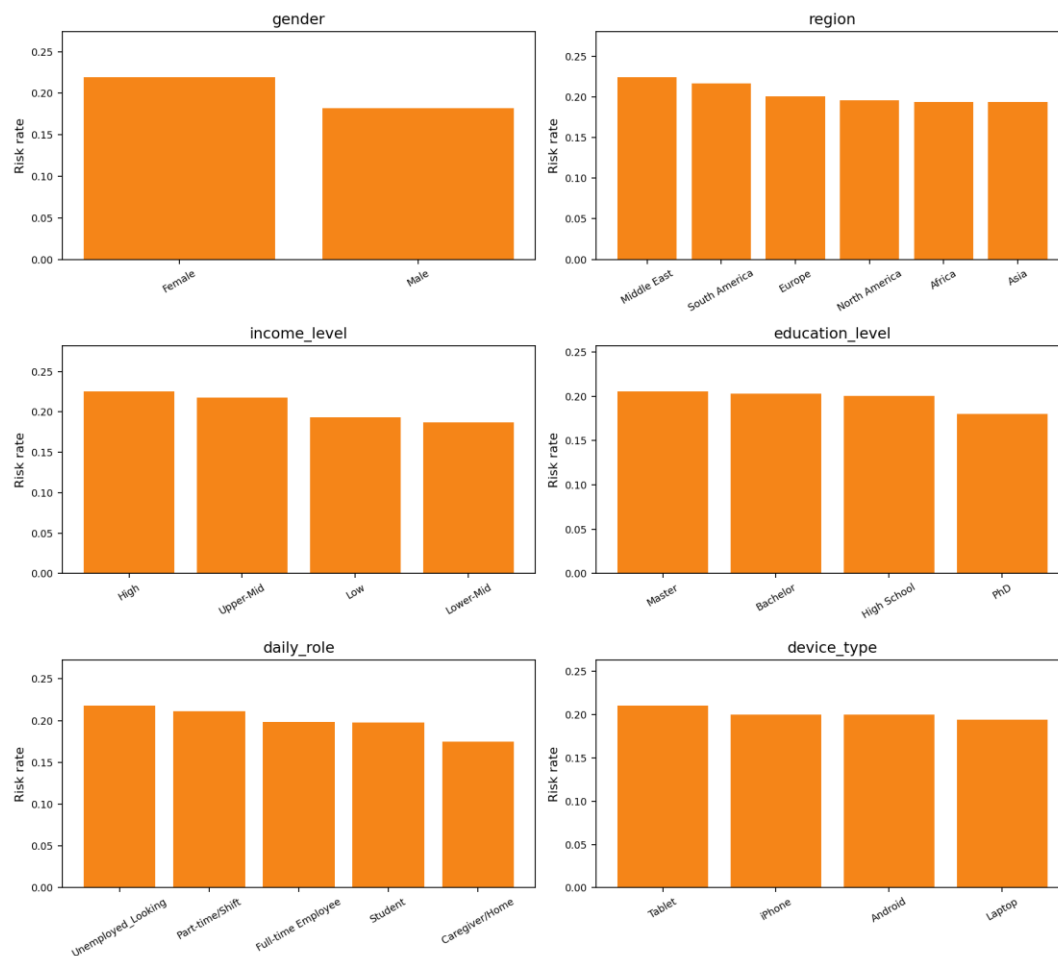


Figure 4-5 Risk rates across categorical variables

The categorical risk-rate bar chart compares risk proportions across variables such as gender, region, income_level, education_level, daily_role, and device_type. This visualization checks whether background categories show systematic descriptive differences, while the main clustering input remains restricted to behavioral and lifestyle numerical variables.

4.6 Relationships between Behavioral Variables and Outcome Variables

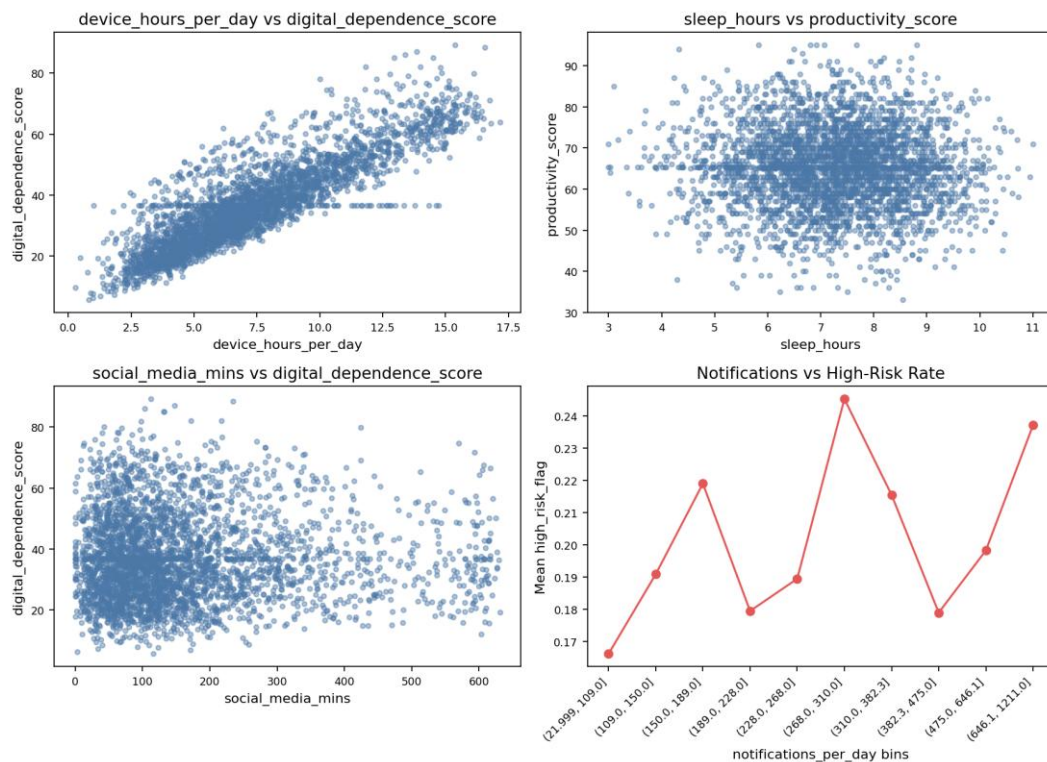


Figure 4-6 Relationships between behavioral variables and outcomes

The scatter plots and binned relationship plots show how selected behavioral variables relate to digital dependence, productivity, and high-risk labels. This satisfies the requirement for scatter-style visualization and supports the transition from exploratory analysis to supervised and unsupervised modeling.

4.7 Summary of EDA Findings

The EDA chapter uses bar charts, histograms, boxplots, scatter plots, and heatmaps with analysis rather than only inserting figures. The visual evidence suggests that digital behavior variables are useful for high-risk screening and digital dependence prediction, while productivity may require additional unobserved factors.

Chapter 5 Machine Learning Modeling, Hyperparameter Tuning, and Evaluation

5.1 Experimental Settings and Evaluation Metrics

All experiments use Python, pandas, numpy, scipy, matplotlib, and scikit-learn on CPU. No GPU, no deep learning, and no unverified experimental results are used. The `random_state` is fixed at 42 for reproducibility. Supervised tasks use train/test splitting, cross-validation, and GridSearchCV-style hyperparameter tuning; the test set is used only for final evaluation.

The classification task reports Accuracy, Precision, Recall, F1, confusion matrix, ROC-AUC, PR-AUC, and Balanced Accuracy. F1 and Recall are emphasized because high-risk screening should reduce missed high-risk cases. The regression task reports R^2 , MSE, RMSE, and MAE. The clustering task uses the elbow method, silhouette coefficient, Calinski-Harabasz, and Davies-Bouldin metrics.

Code 5-1 Cross-validation and GridSearchCV for supervised tuning

```
cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)

search = GridSearchCV(
    estimator=pipeline,
    param_grid=param_grid,
    scoring="average_precision",
    cv=cv,
    n_jobs=-1,
)

search.fit(X_train, y_train)

best_model = search.best_estimator_
```

5.2 Classification Task: High-Risk Digital Lifestyle Screening

The classification target is `high_risk_flag`. Logistic Regression, Random Forest, and Gradient Boosting were evaluated. The final strategy is Gradient Boosting with `threshold=0.14`. This threshold is selected as a recall-oriented screening strategy, not as a medical diagnosis tool.

Table 5-1 Test-set classification metrics under threshold strategies

Model	Threshold strategy	Threshold	Precision	Recall	F1	ROC-AUC	PR-AUC	Balanced Accuracy
Gradient Boosting	default	0.5000	0.6812	0.2670	0.3837	0.7531	0.5084	0.6178
	0.50							

Gradient Boosting	max F1	0.3500	0.5957	0.4773	0.5300	0.7531	0.5084	0.6979
Gradient Boosting	recall $\geq$ 0.60	0.1400	0.4593	0.6420	0.5355	0.7531	0.5084	0.7259
Gradient Boosting	recall $\geq$ 0.70	0.1200	0.2536	0.7955	0.3846	0.7531	0.5084	0.6030

Table 5-2 Validation-set threshold tuning results

Model	Threshold strategy	Threshold	Precision	Recall	F1	PR-AUC	Balanced Accuracy
Gradient Boosting	default 0.50	0.5000	0.6429	0.2045	0.3103	0.4805	0.5880
Gradient Boosting	max F1	0.3500	0.5752	0.4924	0.5306	0.4805	0.7005
Gradient Boosting	recall $\geq$ 0.60	0.1400	0.4175	0.6136	0.4969	0.4805	0.6992
Gradient Boosting	recall $\geq$ 0.70	0.1200	0.2636	0.7727	0.3931	0.4805	0.6149

The final test-set results are Recall=0.6420, F1=0.5355, and PR-AUC=0.5084. The threshold tuning table shows why the default 0.50 threshold is not the only reasonable choice. For high-risk screening, a lower threshold can improve recall, but it also increases false positives.



Figure 5-1 Comparison of tuned classification metrics

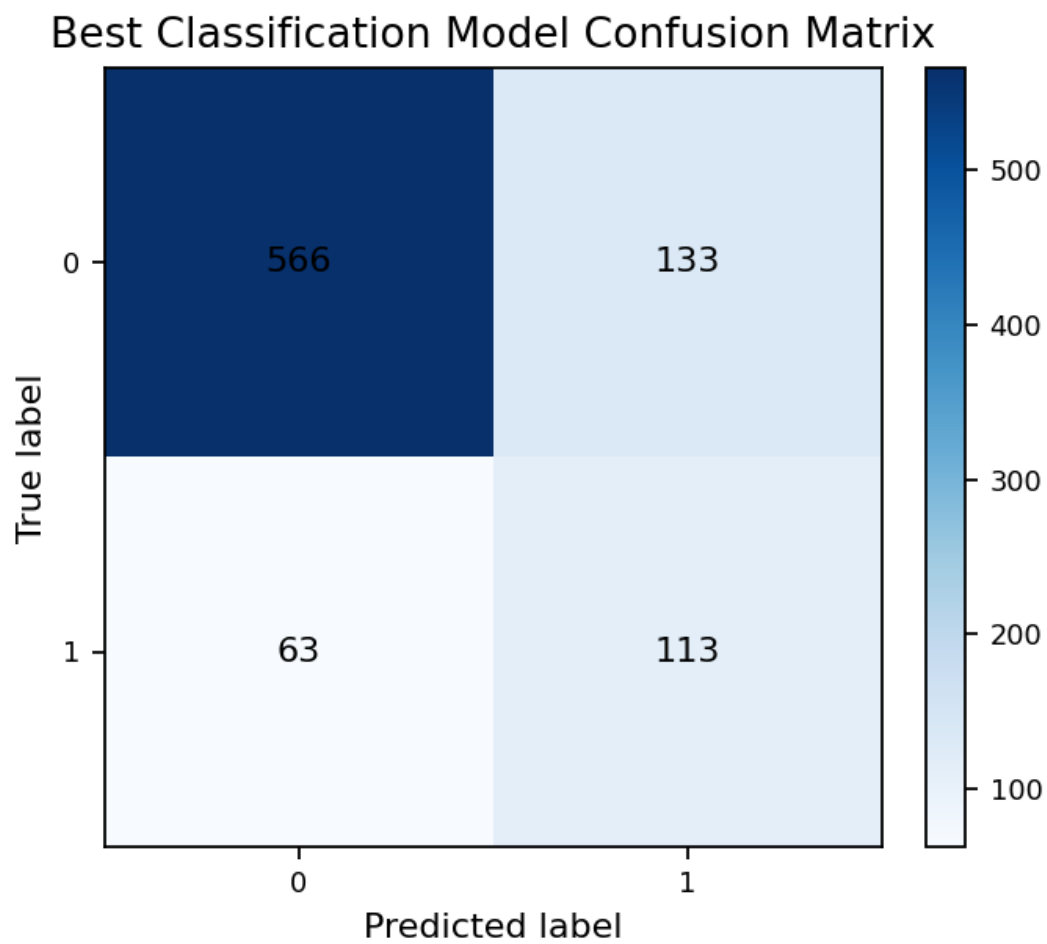


Figure 5-2 Final confusion matrix for high-risk screening

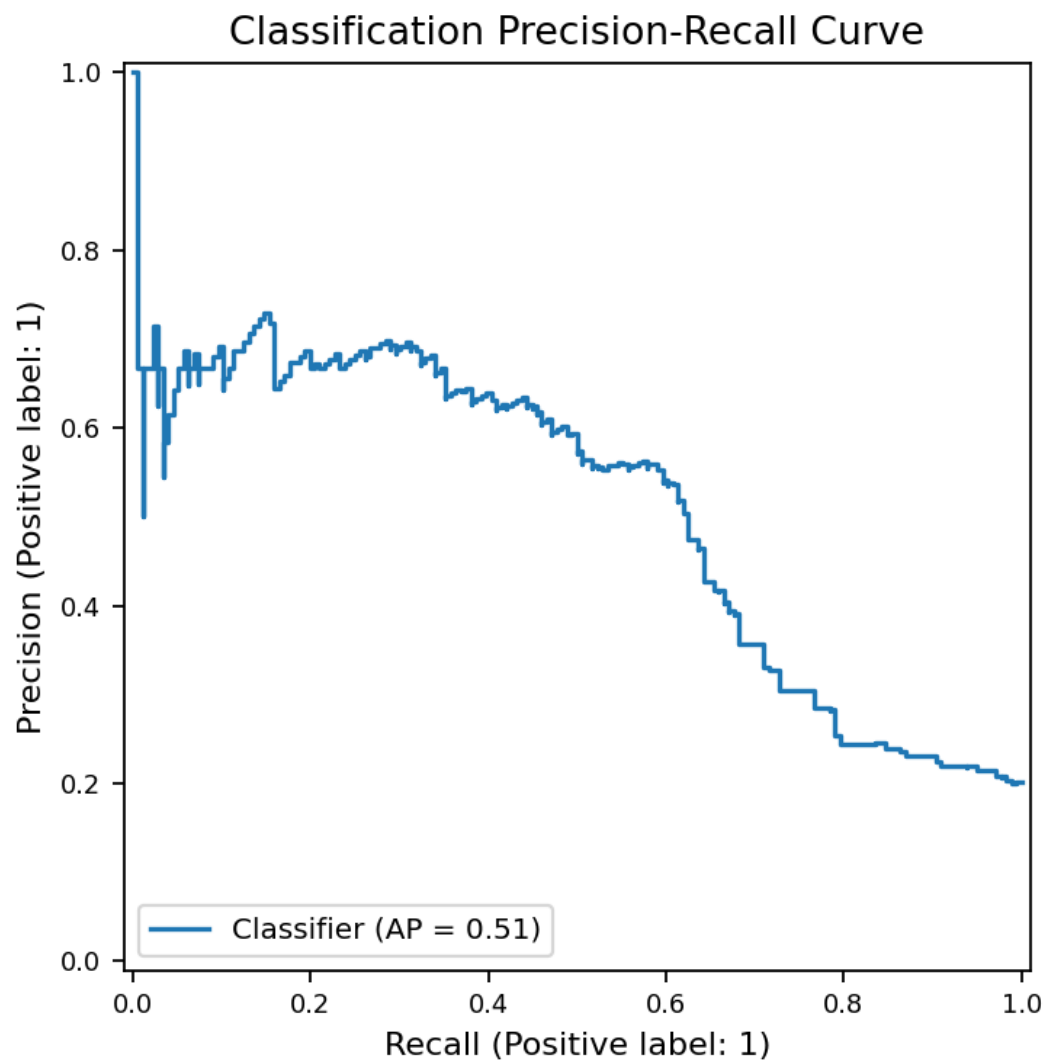


Figure 5-3 Precision-recall curve for the final classifier

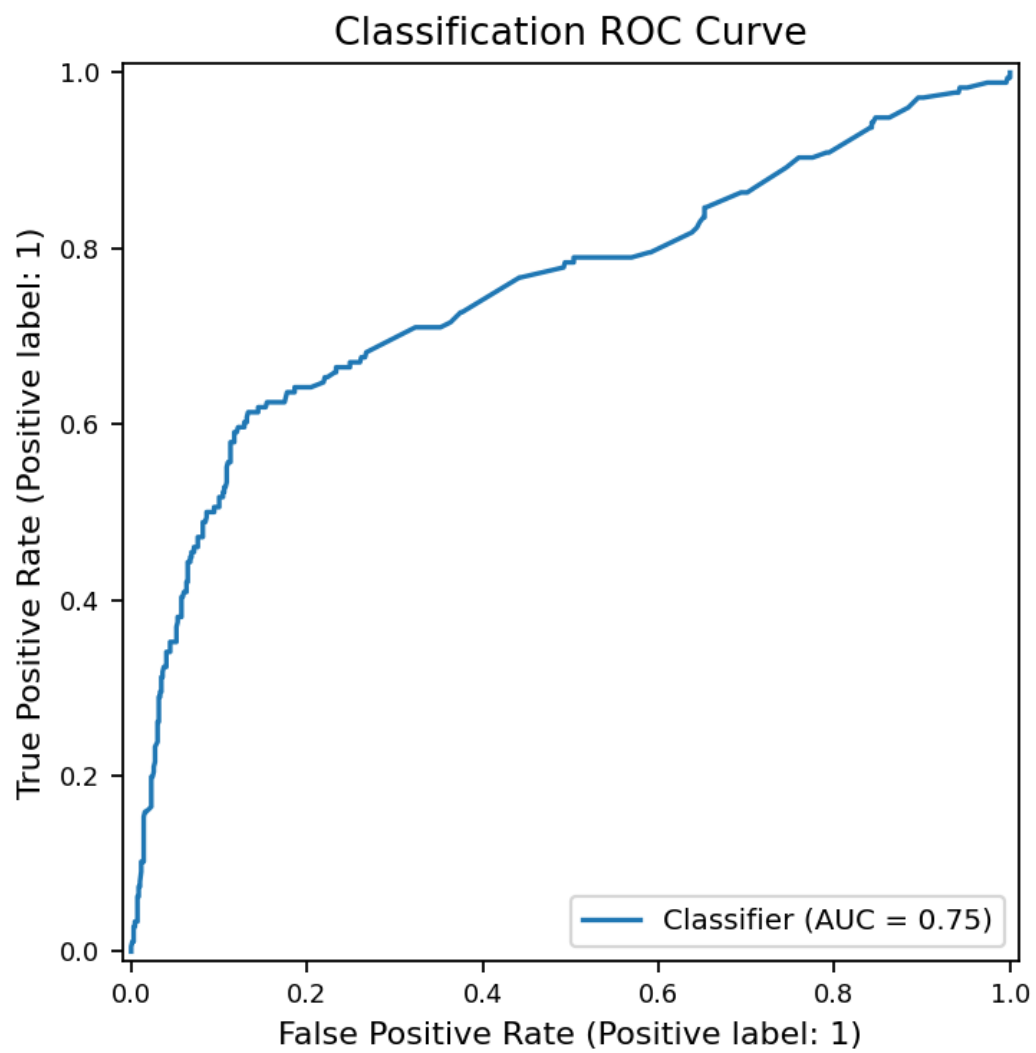


Figure 5-4 ROC curve for the final classifier

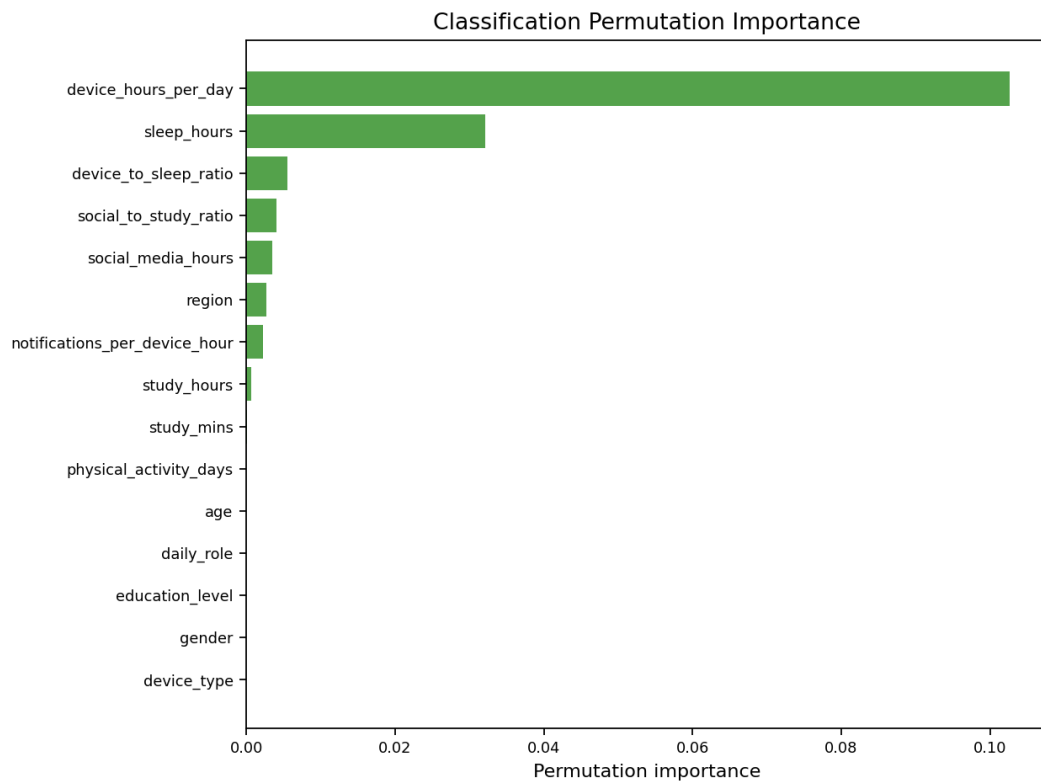


Figure 5-5 Permutation importance for high-risk screening

Modeling results are supported by visual evidence, including the confusion matrix, ROC curve, precision-recall curve, tuned-metric comparison, and permutation importance. The confusion matrix directly shows true positives, false positives, false negatives, and true negatives, while the PR curve is especially useful under class imbalance.

Code 5-2 Threshold selection for a recall-oriented screening model

```
thresholds = np.linspace(0.01, 0.99, 99)

records = []

for threshold in thresholds:

    y_pred = (valid_proba >= threshold).astype(int)

    records.append({

        "threshold": threshold,

        "precision": precision_score(y_valid, y_pred, zero_division=0),

        "recall": recall_score(y_valid, y_pred, zero_division=0),

        "f1": f1_score(y_valid, y_pred, zero_division=0),

    })
```

```
threshold_table = pd.DataFrame(records)

selected = threshold_table.query("recall >= 0.60").sort_values(

    ["precision", "f1"], ascending=False

).iloc[0]
```

5.3 Regression Task: Digital Dependence Prediction and Productivity Weak-Prediction Analysis

The regression main target is digital_dependence_score. The best model obtains $R^2=0.9839$, $MSE=3.1471$, and $MAE=0.9982$. This indicates that digital behavior variables strongly represent digital dependence in this synthetic dataset. However, strong prediction does not imply causation.

Table 5-3 Best-model comparison for the two regression targets

Target variable	Best model	MAE	MSE	RMSE	R^2	Best CV R^2
productivity_score	Gradient Boosting	7.1671	85.2031	9.2306	-0.0041	0.0119
digital_dependence_score	Gradient Boosting	0.9982	3.1471	1.7740	0.9839	0.9804

Table 5-4 Test-set metrics for digital_dependence_score regression

Model	MAE	MSE	RMSE	R^2
Gradient Boosting	0.9982	3.1471	1.7740	0.9839
Ridge	0.7652	3.6541	1.9116	0.9813
Linear Regression	0.7651	3.6538	1.9115	0.9813
Random Forest	1.2680	4.3875	2.0946	0.9776

Table 5-5 Test-set metrics for productivity_score regression

Model	MAE	MSE	RMSE	R^2
Gradient Boosting	7.1671	85.2031	9.2306	-0.0041
Random Forest	7.2088	85.9208	9.2693	-0.0125
Ridge	7.1747	85.5884	9.2514	-0.0086
Linear Regression	7.1999	85.8232	9.2641	-0.0114

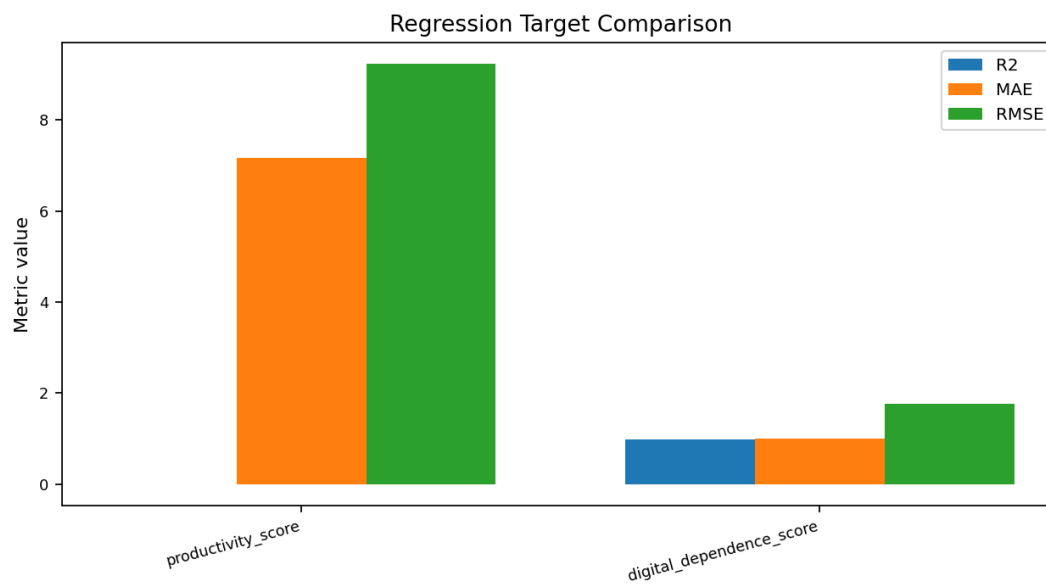


Figure 5-6 Comparison of regression targets

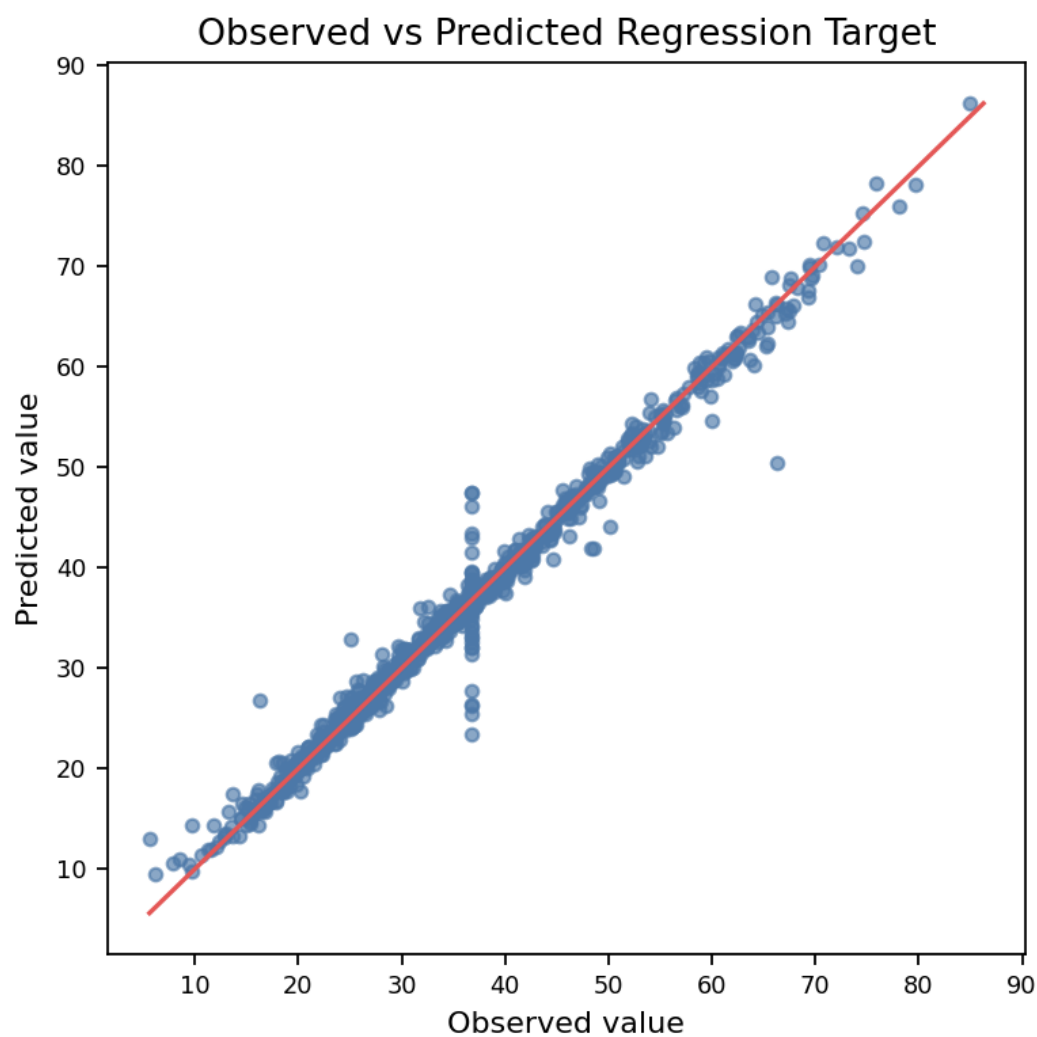


Figure 5-7 Observed versus predicted digital dependence scores

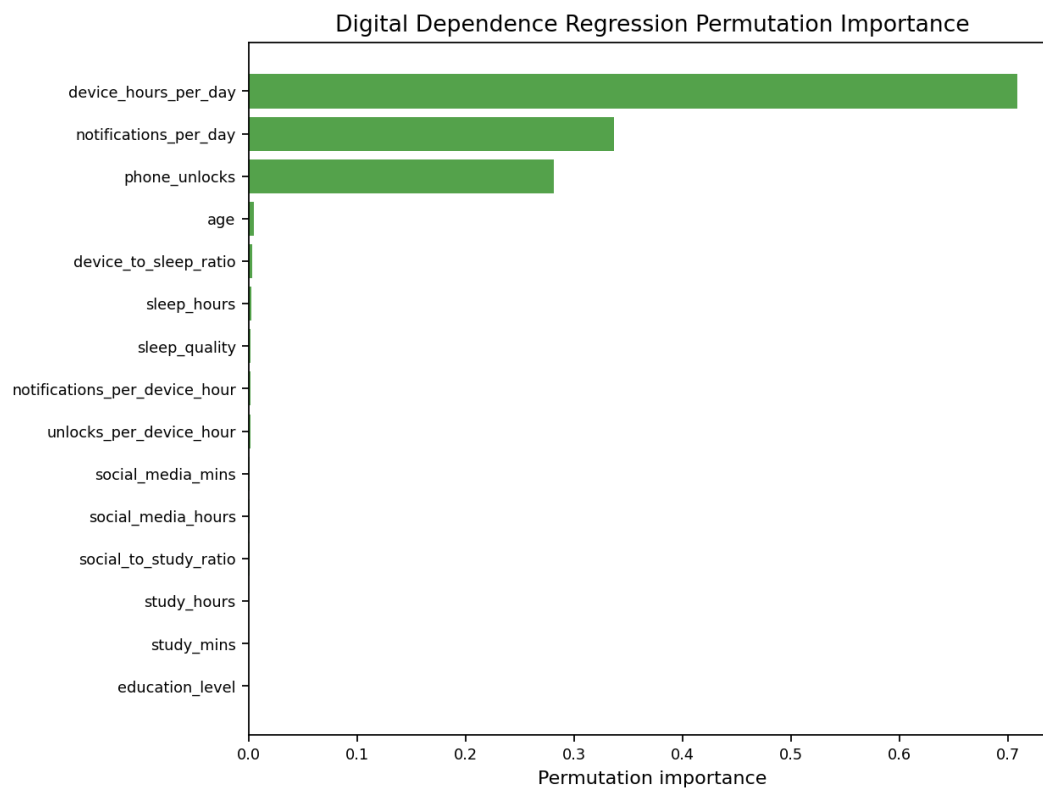


Figure 5-8 Permutation importance for digital dependence prediction

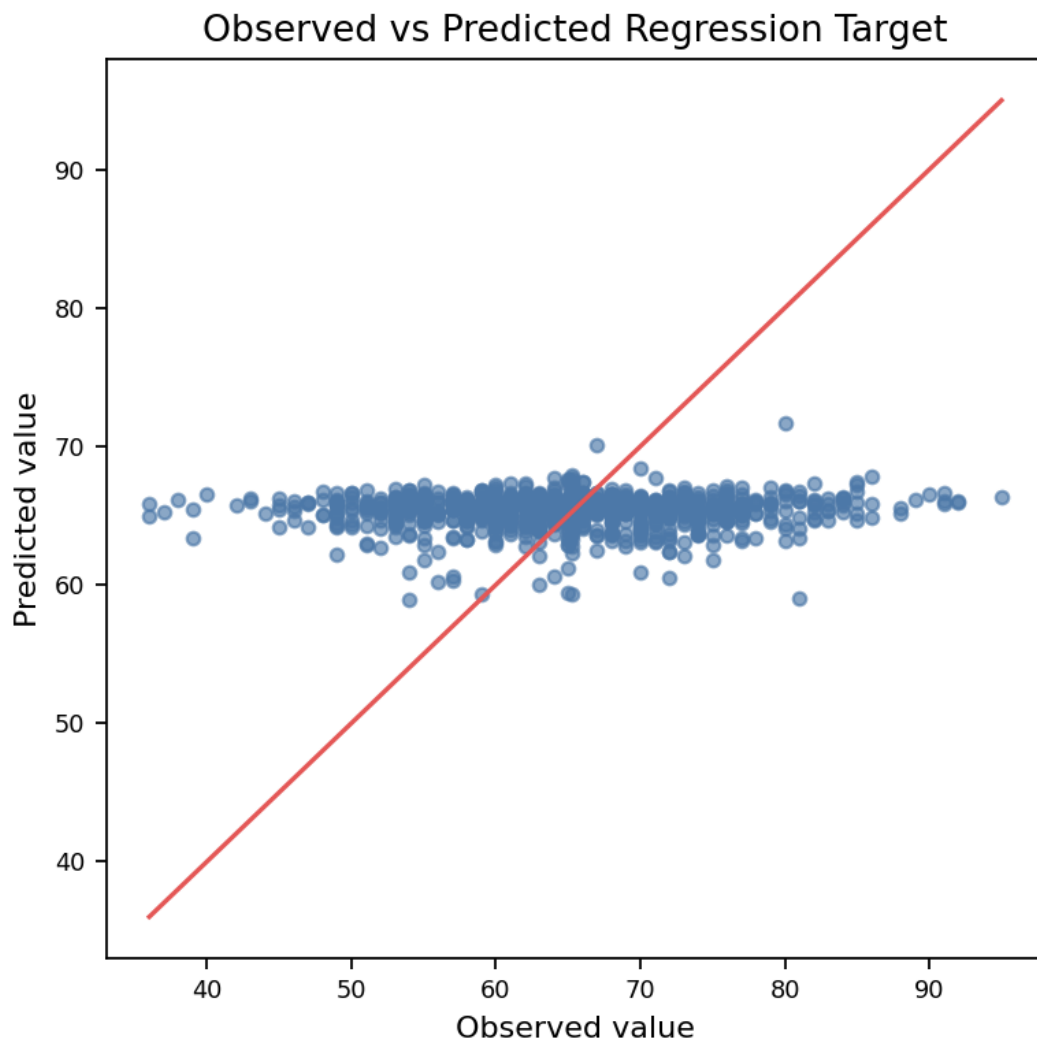


Figure 5-9 Observed versus predicted productivity scores

The productivity_score experiment is deliberately kept as a weak-prediction or negative-result analysis. Its best R^2 is -0.0041, so this report does not claim that productivity can be effectively predicted by the current feature set. This negative result is valuable because it shows that not every target is equally explainable by the same observable variables.

Code 5-3 Regression metric calculation

```
metrics = {  
    "R2": r2_score(y_test, y_pred),  
    "MSE": mean_squared_error(y_test, y_pred),  
    "RMSE": mean_squared_error(y_test, y_pred, squared=False),  
    "MAE": mean_absolute_error(y_test, y_pred),  
}  
  
pd.DataFrame([metrics]).to_csv(output_path, index=False)
```

5.4 Clustering Task: Digital Lifestyle User Profiling

The clustering task uses only digital behavior and lifestyle numerical features. KMeans, AgglomerativeClustering, and GaussianMixture were compared. The elbow method and silhouette coefficient were used for k selection. The best setting is KMeans with k=3 and Silhouette=0.1860.

Table 5-6 Comparison of clustering algorithms and cluster numbers

Algorithm	k	Silhouette	Calinski-Harabasz	Davies-Bouldin
KMeans	2	0.1808	752.8023	1.9065
KMeans	3	0.1860	611.1770	1.7959
KMeans	4	0.1406	546.4364	1.9615
Agglomerative	3	0.1518	495.8938	1.8768
GaussianMixture	2	0.1478	650.8461	2.1599
GaussianMixture	4	0.1225	408.1725	2.2266

Table 5-7 Compact digital lifestyle cluster profiles

Cluster	Cluster size	Device hours	Social media minutes	Sleep hours	Sleep quality	High-risk ratio	Digital dependence score	Suggested profile name
Cluster 0	447	6.9533	430.2125	7.3816	2.7899	0.1969	34.3296	High social-media-use profile
Cluster 1	1039	11.0538	131.8787	6.1218	1.6953	0.3725	51.2425	High device-dependence profile
Cluster 2	2014	5.4711	113.4275	7.8106	3.2137	0.1142	29.6962	Low-load balanced profile

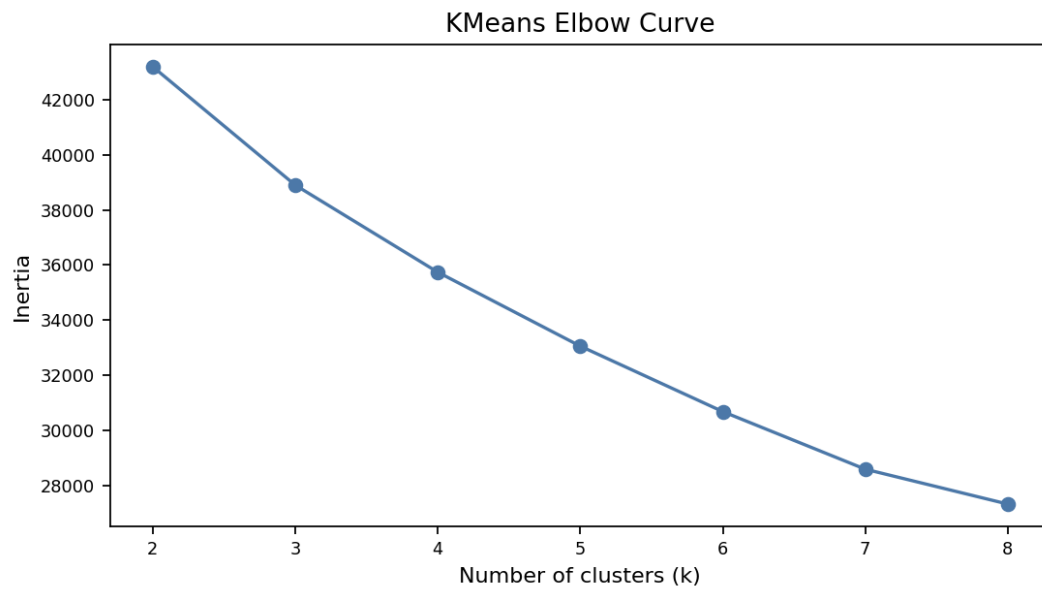


Figure 5-10 KMeans elbow curve for selecting k

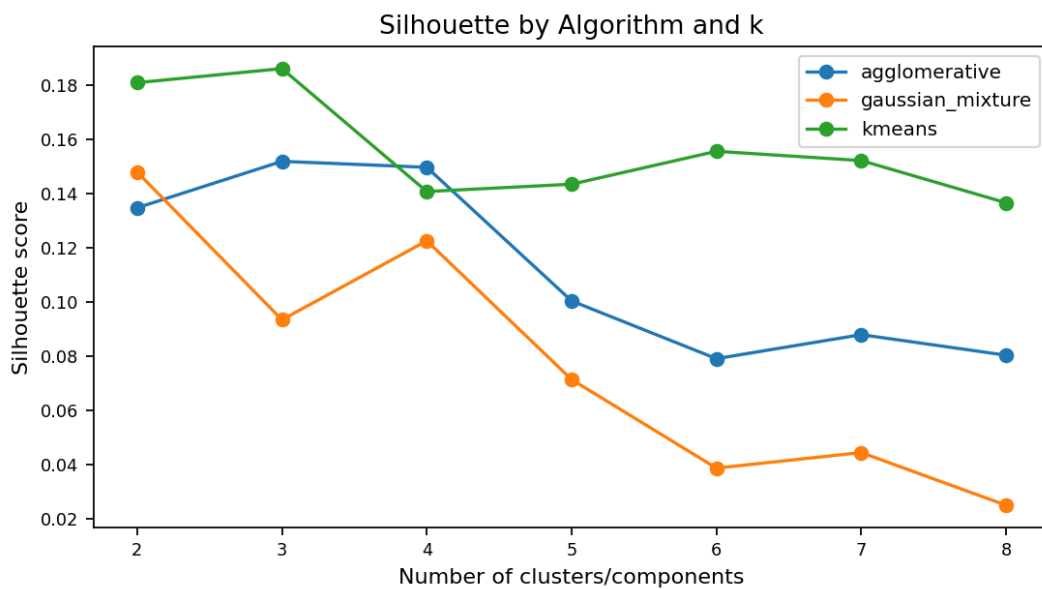


Figure 5-11 Silhouette scores by number of clusters

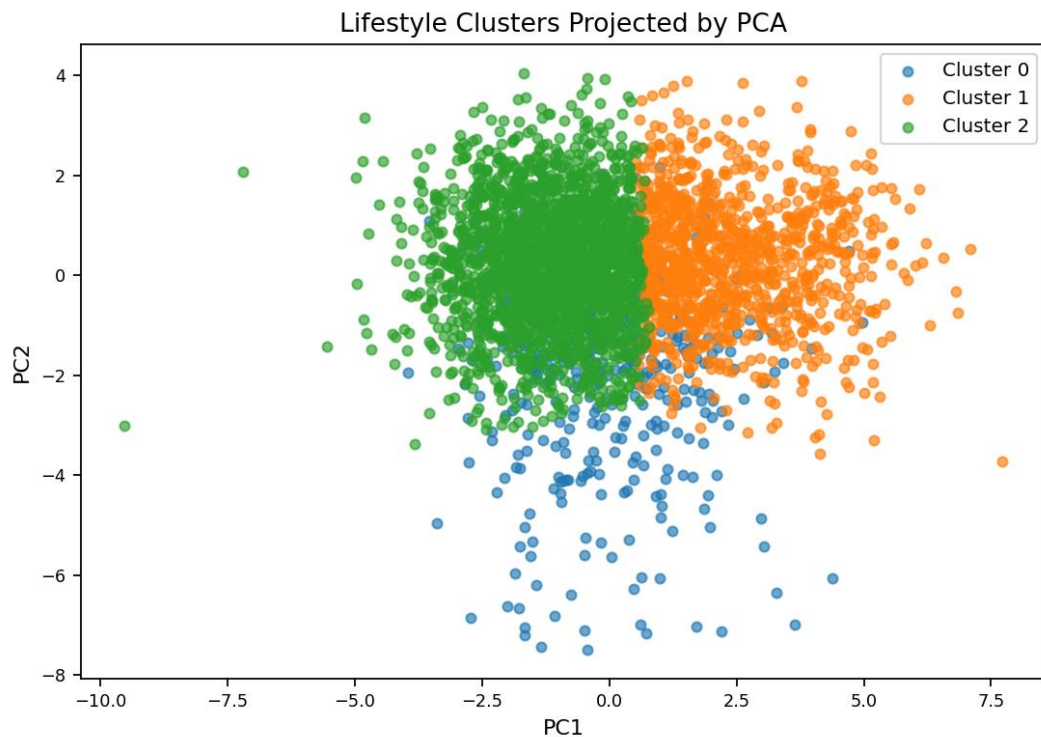


Figure 5-12 PCA visualization of lifestyle clusters

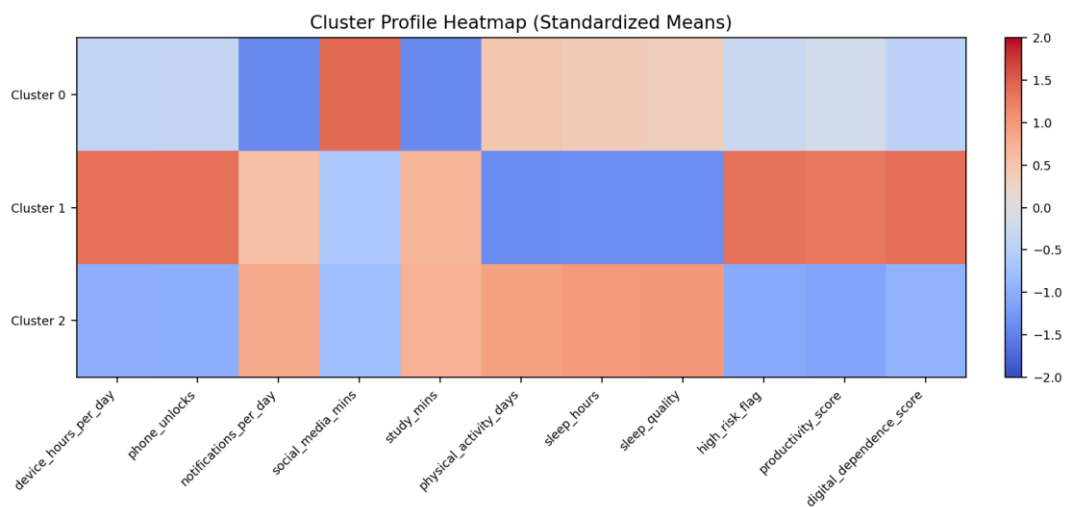


Figure 5-13 Heatmap of cluster lifestyle profiles

The cluster labels are interpreted as high social-media-use profile, high device-dependence profile, and low-load balanced profile. Because Silhouette=0.1860 is relatively low, the clusters should be treated as exploratory profiles rather than naturally clear population boundaries.

Code 5-4 Clustering input scaling and model comparison

```
cluster_X = df[clustering_features]

cluster_X_scaled = StandardScaler().fit_transform(cluster_X)
```

```
for k in range(2, 9):  
    labels = KMeans(n_clusters=k, random_state=42, n_init=20).fit_predict(cluster_X_scaled)  
    scores.append({  
        "algorithm": "KMeans",  
        "k": k,  
        "silhouette": silhouette_score(cluster_X_scaled, labels),  
        "calinski_harabasz": calinski_harabasz_score(cluster_X_scaled, labels),  
        "davies_bouldin": davies_bouldin_score(cluster_X_scaled, labels),  
    })
```

5.5 Integrated Model Result Analysis

The three task types answer different parts of the research question. Classification provides a recall-oriented high-risk screening baseline. Regression shows that `digital_dependence_score` is strongly predictable in the synthetic dataset, while `productivity_score` is weakly predictable. Clustering provides interpretable but weak-boundary lifestyle profiles.

5.6 Business Interpretation and Result Boundaries

The results can be used as a reference for digital lifestyle management, notification control, social media time awareness, sleep and activity balance, and manual review of high-risk screening results. They should not be used for medical diagnosis, automatic punishment, or strong intervention. The synthetic benchmark nature of the dataset limits direct real-world generalization.

Chapter 6 Report Structure and Code Presentation

6.1 Project Directory and Engineering Structure

The report follows a standard data analysis workflow: define analytical objectives, introduce the dataset, perform preprocessing, conduct statistical analysis and visualization, build and tune machine learning models, evaluate results, and summarize limitations and reflections.

Table 6-1 Project directory and code organization

Directory/File	Role in the report
data/raw and data/processed	Store raw and processed dataset files without deleting the original data.
notebooks/00-05	Record dataset selection, preprocessing/EDA, classification, regression, clustering, and result summary.
src/config.py	Centralize paths, random_state=42, target names, and leakage-column definitions.
src/data_utils.py	Provide dataset loading, saving, and reusable quality-check helpers.
src/feature_engineering.py	Generate engineered behavior-ratio and interaction features and define task-specific feature sets.
src/model_utils.py	Provide model evaluation, metric calculation, threshold tuning, and clustering utilities.
scripts/stage2_5_evidence_enhancement.py	Generate preprocessing quality checks, PCA explained variance evidence, and final report artifact lists.
results/ and figures/	Store CSV metrics, summaries, tables, and all visual evidence used in the report.
final_submit/	Store the final Word submission file.

6.2 Notebook and Python Source Code Organization

The notebooks are executed in the order 00_dataset_selection_and_compliance, 01_data_preprocessing_and_eda, 02_classification_high_risk, 03_regression_productivity, 04_clustering_lifestyle_profiles, and 05_result_summary_for_report. The source files include config.py, data_utils.py, feature_engineering.py, model_utils.py, and visualization.py.

6.3 Core Code Snippets and Explanations

The main text embeds core code snippets with explanations, while complete code is provided in notebooks/, src/, and scripts/. This balances readability and reproducibility: readers can understand key logic in the report and still inspect the full implementation in the project directory.

Code 6-1 Reproducible configuration

```
RANDOM_STATE = 42

CLASSIFICATION_TARGET = "high_risk_flag"

REGRESSION_TARGETS = [

    "productivity_score",

    "digital_dependence_score",

]

HIGH_RISK_LEAKAGE_COLUMNS = [

    "anxiety_score", "depression_score", "stress_level",

    "happiness_score", "focus_score", "productivity_score",

    "digital_dependence_score",

]
```

6.4 Figures, Result Files, and Reproducibility

All major results are saved as CSV files in results/ and all visual evidence is saved as PNG files in figures/. This makes the report auditable: every metric and figure can be traced back to a generated artifact.

Code 6-2 Artifact-oriented result saving

```
metrics_df.to_csv(results_dir / "classification_tuned_metrics.csv", index=False)

threshold_df.to_csv(results_dir / "classification_threshold_tuning.csv", index=False)

profile_df.to_csv(results_dir / "clustering_lifestyle_profiles_compact.csv", index=False)

fig.savefig(figures_dir / "classification_precision_recall_curve.png", dpi=160)

fig.savefig(figures_dir / "clustering_lifestyle_profile_heatmap.png", dpi=160)
```

6.5 Complete Code Appendix Description

The complete code is not fully copied into the main text because long notebooks would reduce readability. Instead, Appendix A explains the core code structure, Appendix B gives the running order, and Appendix C lists important result files and figures.

6.6 GitHub Repository, Project Structure, and Static Showcase

The complete experimental code, notebooks, generated figures, result CSV files, and final report materials are organized in the GitHub repository below. The static showcase page is a lightweight browser-based presentation of the project workflow and visual results.

Artifact	Location
GitHub repository	https://github.com/hansu650/Big-Data-Homework
Project directory	Big-Data-Homework / final project directory
Final report	final_submit / final Word course report
Overleaf/LaTeX package	overleaf_final
Static visualization	web_demo / index.html
Optional HTML preview	https://htmlpreview.github.io/?https://github.com/hansu650/Big-Data-Homework/blob/main/%E6%9C%9F%E6%9C%AB%E8%80%83%E6%9F%A5%E6%8A%A5%E5%91%8A_%E6%95%B0%E5%AD%97%E7%94%9F%E6%B4%BB%E6%96%B9%E5%BC%8F%E5%88%86%E6%9E%90/web_demo/index.html

Project structure screenshot

Project Directory and Engineering Structure

Big-Data-Homework / final project directory

```
final project directory/  
|-- data/  
|-- notebooks/  
|-- src/  
|-- scripts/  
|-- figures/  
|-- results/  
|-- final_submit/  
|-- overleaf_final/  
|-- web_demo/  
`-- requirements.txt
```

Figure 6-1 Project directory and engineering structure

The project is organized around data, notebooks, source code, scripts, figures, results, and final submission artifacts.

This structure demonstrates that the report is supported by reproducible engineering materials rather than only by a standalone document.

Notebook and source-code screenshot

Notebook and Source-Code Organization

Big-Data-Homework / final project directory

```
notebooks/  
|-- 00_dataset_selection_and_compliance.ipynb  
|-- 01_data_preprocessing_and_eda.ipynb  
|-- 02_classification_high_risk.ipynb  
|-- 03_regression_productivity.ipynb  
|-- 04_clustering_lifestyle_profiles.ipynb  
`-- 05_result_summary_for_report.ipynb  
  
src/  
|-- config.py  
|-- data_utils.py  
|-- feature_engineering.py  
|-- model_utils.py  
`-- visualization.py
```

Figure 6-2 Notebook and source-code organization

The notebook sequence follows the course-report workflow from dataset compliance checking to final result summary.

The src directory separates configuration, data utilities, feature engineering, model utilities, and visualization helpers.

Chapter 7 Conclusions and Personal Reflection

7.1 Core Experimental Conclusions

This report completed independent dataset selection, data preprocessing, EDA, classification, regression, clustering, result visualization, code organization, and final Word report preparation. The classification task provides a recall-oriented screening model; the regression task shows strong predictability for `digital_dependence_score` and weak predictability for `productivity_score`; the clustering task provides exploratory lifestyle profiles.

7.2 Data Patterns and Practical Value

The descriptive and modeling results suggest that high device usage, notifications, and social media behavior are associated with high-risk labels and digital dependence patterns in this synthetic dataset. These patterns can support digital lifestyle awareness, notification management, sleep/activity balance, and manual review of screening results.

7.3 Model Advantages, Limitations, and Risks

The classification model improves recall by lowering the threshold, but this also increases false positives. The regression model for digital dependence is strong, but it does not prove causality. The productivity regression result is weak, which shows that the current variables do not explain every outcome. The clustering profiles are interpretable, but the silhouette score is low, so the boundaries are not strong.

7.4 Future Improvements

Future work could use real-world longitudinal data, richer behavioral features, external validation, more robust threshold-selection protocols, and additional unsupervised modeling methods. If privacy and ethics conditions are satisfied, time-series behavioral features may also improve the interpretation of digital lifestyle changes.

7.5 Course Learning Reflection and Appreciation

During the course, the teacher repeatedly emphasized that a qualified big data analysis report should not only present model scores, but also explain data cleaning, feature extraction, visualization, evaluation metrics, analytical conclusions, and reproducibility. These reminders helped me organize the whole project in a more systematic way.

Through this final report, I learned that data compliance, target leakage control, metric selection, negative-result interpretation, visualized evidence, and code organization are all essential parts of a complete analysis. I sincerely appreciate the teacher's clear teaching, patient guidance, and emphasis on rigorous experimental reporting throughout the course.

References

- [1] Digital Lifestyle Benchmark Dataset (2025). Public dataset page and data card.
- [2] scikit-learn official documentation. <https://scikit-learn.org/>
- [3] Pedregosa F., Varoquaux G., Gramfort A., et al. Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 2011.
- [4] pandas official documentation. <https://pandas.pydata.org/>
- [5] NumPy official documentation. <https://numpy.org/>
- [6] Matplotlib official documentation. <https://matplotlib.org/>
- [7] Course requirements for Big Data Analysis and Applications final course report.

Appendix A Core Code

Appendix A records the purpose of the core source files and the main code snippets already shown in the report. The complete executable code is located in notebooks/, src/, and scripts/. The report avoids copying every notebook cell into the main text so that the reading flow remains clear.

Appendix B Full Experimental Running Instructions

Install dependencies with `pip install -r requirements.txt`.

Run notebooks in the order 00 to 05.

Run `scripts/stage2_5_evidence_enhancement.py` when preprocessing quality checks and PCA evidence need to be regenerated.

Use CPU only and keep `random_state=42`.

Open the final DOCX in Word/WPS, fill personal information, check pages, and export PDF manually if the local converter is unavailable.

Appendix B Repository Access and Local Opening Notes

The complete repository is available at <https://github.com/hansu650/Big-Data-Homework>.

The final project directory contains notebooks, source code, scripts, figures, results, final submission files, the LaTeX package, and the static showcase page.

To inspect the static showcase, open `web_demo/index.html` directly in a browser.

Step	Local command or action
Install environment	<code>pip install -r requirements.txt</code>
Run notebooks	Run notebooks 00 to 05 in order
Open final report	Open <code>final_submit</code> / final Word course report
Open showcase	Open <code>web_demo</code> / <code>index.html</code> in a browser

Appendix C Result Files and Figure List

Important result files include `dataset_compliance_summary.csv`, `preprocessing_quality_check.csv`, `classification_tuned_metrics.csv`, `classification_threshold_tuning.csv`, `regression_target_comparison.csv`, `regression_digital_dependence_metrics.csv`, `regression_productivity_metrics.csv`, `clustering_model_comparison.csv`, `clustering_lifestyle_profiles_compact.csv`, and `pca_explained_variance.csv`.

Table 10-1 Result file and figure categories

Artifact group	Examples
EDA figures	target distribution, histograms, correlation heatmap, boxplots, categorical risk-rate bars, scatter plots
Classification figures	tuned metrics comparison, confusion matrix, PR curve, ROC curve, permutation importance
Regression figures	target comparison, observed-vs-predicted plots, permutation importance
Clustering/PCA figures	elbow curve, silhouette curve, PCA explained variance, PCA cluster plot, profile heatmap

Appendix C Static Visualization Showcase

GitHub repository: <https://github.com/hansu650/Big-Data-Homework>

Static visualization showcase: web_demo/index.html

Optional HTML preview:

https://htmlpreview.github.io/?https://github.com/hansu650/Big-Data-Homework/blob/main/%E6%9C%9F%E6%9C%AB%E8%80%83%E6%9F%A5%E6%8A%A5%E5%91%8A_%E6%95%B0%E5%AD%97%E7%94%9F%E6%B4%BB%E6%96%B9%E5%BC%8F%E5%88%86%E6%9E%90/web_demo/index.html

The static page can be opened locally in a browser and does not require a server, GPU, checkpoint, or online inference.

Showcase file	Purpose
web_demo/index.html	Static overview dashboard and visualization gallery
web_demo/README_WEB_DEMO.md	Local opening instructions and boundaries
web_demo/assets/figures/	Copied report figures used by the gallery
web_demo/assets/screenshots/	Screenshots inserted into the Word report

Static showcase homepage

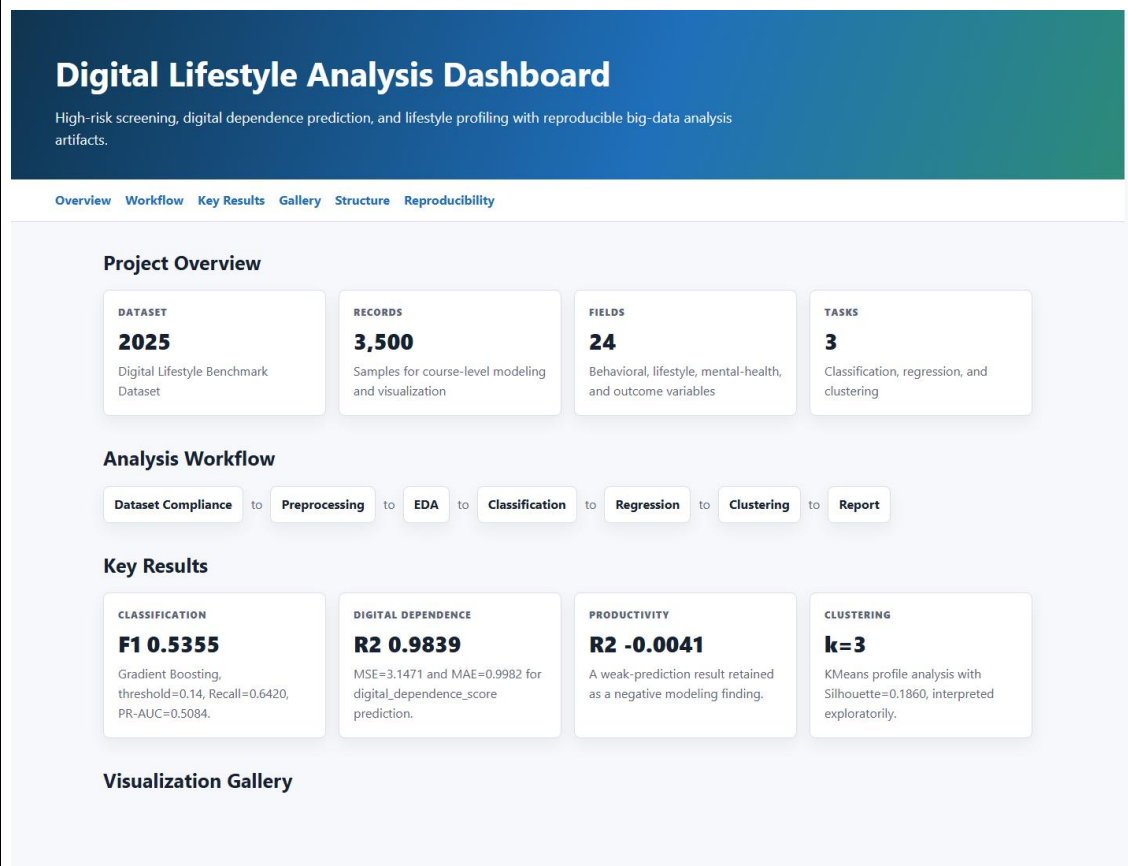


Figure C-1 Static visualization showcase homepage

The homepage summarizes the project theme, workflow, dataset scale, task types, and key modeling results.

It provides a concise visual entry point for readers who want to inspect the project before reading the full report.

Static showcase gallery



Figure C-2 Visualization gallery in the static showcase page

The gallery collects representative EDA, classification, regression, and clustering figures from the project artifacts.

It is intended for static display only and does not perform online prediction or model inference.

Appendix D Personal Contribution and Academic Integrity Statement

This report was independently completed for the Big Data Analysis and Applications course. Previous experiments were used only as workflow references. The final topic, dataset selection, feature design, EDA, modeling tasks, visualizations, interpretations, Word formatting, and conclusions were redesigned for this final course report.

The report does not copy the artificial intelligence course report content. Personal information remains blank and should be filled manually before final submission.